Chapter 9
Investigations on Chemical, Mechanical, and Long-Term Characteristics of Alkali-Activated Concrete

Arkamitra Kar, Kruthi Kiran Ramagiri, Sriman Pankaj Boindala, Indrajit Ray, Udaya B. Halabe, and Avinash Unnikrishnan

9.1 Introduction

Rapid global development of infrastructure has resulted in the consumption of 4.1 billion tons (BTs) of limestone-based portland cement (PC), with an anticipated increase to 4.83 BTs by 2030 (Garside 2021). This accounts for 5–8% of the total global CO_2 emission, as the 1 kg of PC leads to nearly 1 kg of CO_2 production. India alone is estimated to produce 0.298 BTs of PC per annum (PA), whereas the USA produces 0.09 BTPA with a steady increase anticipated through the decade (BEEINDIA 2021). It is also reported that India produces an estimated 0.225 BTPA

A. Kar (✉) · K. K. Ramagiri
Department of Civil Engineering, BITS-Pilani, Hyderabad Campus, Hyderabad 500078, Telangana, India
e-mail: arkamitra.kar@hyderabad.bits-pilani.ac.in

K. K. Ramagiri
e-mail: p20170008@hyderabad.bits-pilani.ac.in

S. P. Boindala
Civil and Environmental Engineering, Technion—Israel Institute of Technology, Haifa, Israel

I. Ray
Department of Civil and Environmental Engineering, University of West Indies, St. Augustine, Trinidad and Tobago
e-mail: Indrajit.Ray@sta.uwi.edu

U. B. Halabe
Department of Civil and Environmental Engineering, West Virginia University, Morgantown, WV 26506, USA
e-mail: udaya.halabe@mail.wvu.edu

A. Unnikrishnan
Department of Civil and Environmental Engineering, Portland State University, Portland, OR 97201, USA
e-mail: uavinash@pdx.edu

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2022

K. R. Reddy et al. (eds.), *Advances in Sustainable Materials and Resilient Infrastructure*, Springer Transactions in Civil and Environmental Engineering,
https://doi.org/10.1007/978-981-16-9744-9_9

of fly ash, with around 30% being discarded as waste and less than 1% in concrete production (Yousuf et al. 2020). In the USA, out of the 0.04 BTPA of fly ash produced, around 55% is reused with a goal of 90% utilization by 2030 (Ghazali et al. 2019). In Europe, almost 80% of the fly ash produced as waste is utilized in construction activities to reduce PC usage (Euorstat 2020). Fly ash is popularly used as a mineral admixture in cementitious systems up to nearly 70% by volume of PC in high-volume applications. More recently, another technique to utilize fly ash is the implementation of alkali-activated concrete (AAC). The corresponding alkali-activated binder (AAB) is developed through the reaction of industrial waste like fly ash as a precursor and an alkaline activator composed of sodium silicate (SS) and sodium hydroxide (SH). The chemical composition of the resulting sodium aluminosilicate matrix is different from the calcium silicate hydrate matrix in traditional hydrated PC paste. The hardened characteristics of AAC aided with thermal curing at 60–80°C are comparable to those of PC concrete (Provis 2018). Presently, AAC is utilized in the form of precast members such as the Toowoomba Wellcamp Airport in Brisbane, Australia, but the implementation of ambient curing can encourage its widespread practical usage (Glasby et al. 2015). To achieve a sustainable practice, it is important to prevent energy consumption due to thermal curing and simultaneously compensate for the slow strength gain due to the lower reaction rate of fly ash. The inclusion of slag to partially replace fly ash is implemented in this regard (Ramagiri et al. 2021).

It is reported that the mechanical strength values of AAC are comparable or sometimes even higher than that of PC concrete (Provis 2018; Ramagiri et al. 2021). However, there is limited information reported on the long-term characteristics of AAC. Increasing global development of infrastructure with greater risk of fire exposure mandates the need for highly fire-resistant concrete. It is reported that thermally cured fly ash-based AAC exposed to severely high temperatures displays higher residual strength and greater resistance to spalling compared to PC concrete. The reduction of mechanical strength of PC concrete upon exposure to 400–600 °C is attributed to the decomposition of its calcium silicate hydrate (C-S-H) matrix and calcium hydroxide ($Ca(OH)_2$) with the loss of chemically bound water, followed by the complete disintegration of the matrix beyond 600 °C. The performance of AAC is comparatively enhanced due to the formation of a pseudo-viscous phase, resulting from the sintering of the fly ash (Martin et al. 2015). The response of AAC to elevated temperatures depends on its precursors and the subsequent formation of a sodium aluminosilicate hydrate (N-A-S-H) polymeric matrix, which is starkly different from the C-S-H in PC (Kar 2013). The associated porosity in hardened PC concrete and AAC are consequently different, thereby directly governing the resistance to elevated temperatures. The selected precursor-activator and curing temperature combination governs the initial mechanical strength of AAC, and subsequently, its residual strength. It is reported that the initial strength and residual strength of fly ash-based AAC are inversely proportional when the exposure temperature is increased to 600 °C. This happens due to further polymerization of the unreacted precursor residue at elevated temperatures, leading to the greater formation of the N-A-S-H polymer matrix. However, in the case of fly ash-slag blended precursor, the N-A-S-H matrix coexists with a calcium aluminosilicate hydrated (C-A-S-H)

matrix formed by the alkali-activation of slag. Hence, the residual strength of blended AAC decreases on exposure to elevated temperatures due to the disintegration of this C-A-S-H, similar to that of the C-S-H in PC-based systems (Ramagiri and Kar 2020).

A critical parameter for the widespread application of reinforced AAC is its bond strength, especially at elevated temperatures caused by fire incidents. The bond strength of concrete is expressed as idealized uniform stress developed at the concrete-steel interface. The bond strength of thermal-cured AAC is governed by its mix proportions, duration of curing, effective cover, size and type of rebar, and development length (Ramagiri et al. 2021; Adak et al. 2017; Castel and Foster 2015; Sarker 2011). Observations from pull-out tests conducted on AAC and PC concrete reveal superior bond strength results in the former, owing to enhanced resistance to tensile stress and a more compact interfacial transition zone (ITZ) (Sarker 2011). In the case of rebars having smaller diameters, enhanced bonding of the rebar and concrete facilitates breaking of the bar over slippage. In contrast, larger bar diameter fails due to concrete splitting (Castel and Foster 2015; Fernandez-Jimenez 2006). The bond strength of AAC reinforced with mild steel or deformed bars is improved in the presence of thermal curing or nanosilica addition (Adak et al. 2017; Castel and Foster 2015). However, limited literature is reported on the characteristics of ambient-cured AAC and its bond behavior at elevated temperatures. Hence, this study focuses on assessing the strength characteristics of ambient-cured AAC through compressive and bond strength tests after exposure to elevated temperatures; and to determine an optimal precursor combination based on these results. Mineralogical, chemical, and morphological analyses are correlated with these findings after validation using existing data (Ramagiri and Kar 2019). The following section presents the materials used for this study and the detailed methodology adopted for the relevant tests conducted.

9.2 Materials and Methods

9.2.1 Materials

Class F fly ash (specific surface 490 m^2/kg; sp. gr. 2.06; loss on ignition, LOI 3%; pozzolanic activity index, PAI 96.46%) conforming to ASTM C618-19 (2019) is supplied by the National Thermal Power Corporation (NTPC) plant in Ramagundam, India. The slag used for this study, conforming to Grade 100 (specific surface 580 m^2/kg; sp. gr. 2.71; LOI 1.41%; PAI 114.46%) of the ASTM C989/C989M-18a (2018), is procured from JSW Cement Ltd. The relevant oxide compositions and particle size information are provided in Table 9.1.

SH used in this study to prepare the alkaline activator conforms to rayon grade with 99% purity. SS solution is composed of 55.1% water, 29.5% SiO_2, and 14.7% Na_2O by weight (to maintain initial silicate modulus of 2:1). Both these components are obtained from HYCHEM laboratories. The liquid blend of crushed SH pellets

Table 9.1 Specification of raw materials

Component (%)	Fly ash	Slag
CaO	1.78	40.64
SiO_2	60.13	35.15
Al_2O_3	28.37	19.60
Fe_2O_3	5.10	0.53
SO_3	0.11	1.89
K_2O	2.16	0.40
TiO_2	1.42	0.92
d_{50} (μm)	51.90	13.93

and SS solution generates heat. So, it is allowed to cool down for 24 h before concrete preparation to alleviate this heat.

Fine aggregate in the form of river sand and coarse aggregate in the form of crushed granite are used to comply with Zone II of IS 383 (2016). The respective water absorption values are 0.5% and 0.1%; the corresponding specific gravity values are 2.65 and 2.72. High-range water-reducing admixture (HRWRA), conforming to Type F of ASTM C494-13 (Specification for Chemical Admixtures for Concrete) is used. For the testing of bond strength, deformed steel bars having 0.2% proof stress of 500 MPa and 12 mm diameter are used.

9.2.2 Mix Proportions and Specimen Preparation

Fly ash: slag ratios in the AAC mixes are varied as 100% fly ash (FS 0), 70% fly ash + 30% slag (FS 30), 60% fly ash + 40% slag (FS 40), and 50% fly ash + 50% slag (FS 50) respectively. The silicate modulus, Ms, for the activator is maintained at 1.4 based on a previous study (Kar et al. 2013). Coarse aggregate quantity is maintained at 1209 kg/m^3 and fine aggregate at 651 kg/m^3 for each mix, respectively. A slump height of 75–100 mm is maintained for each mix by maintaining water to total solids (w/s) ratio of 0.3. The extra water required to maintain w/s is calculated after taking into account the water present in the SS solution (Table 9.2).

During mixing, a uniform blend of the dry precursors with the coarse aggregates and fine aggregates is ensured. Then, the preblended activator is added to the dry mix. After mixing for two to three minutes, previously calculated additional water is added, followed by the required HRWRA to achieve the desired workability. FS 0 specimens, being solely based on the lower activation of fly ash under ambient conditions, are kept in sealed condition and demolded after a week to allow for full hardening and to prevent any damage to the specimens. The other blended AAC mixes are demolded after 48 h and moist-cured for a week. Then, they are stored at ambient laboratory conditions (31 ± 2 °C average temperature and 70% average

Table 9.2 AAC mix proportions in (kg/m^3)

Mix components	FS 0	FS 30	FS 40	FS 50
Fly ash	400	280	240	200
Slag	0	120	160	200
SH	10.57	10.57	10.57	10.57
SS solution	129.43	129.43	129.43	129.43
Extra water	77.38	77.38	77.38	77.38
HRWRA (l/m^3)	0	3.14	3.57	4

relative humidity) prior to testing. For this study, all specimens are ambient-cured, and triplicates are tested at the age of 28 days, as elaborated below.

9.2.3 Experimental Methods

The time–temperature curves presented in ASTM E119-20 (2020) are used to select exposure temperatures of 31 ± 2 °C (ambient), 538 °C, 760 °C, and 892 °C for the AAC specimens. Using an AIMIL muffle furnace, the specimens are exposed to these environments 2 h prior to the mechanical strength tests to attain thermal equilibrium with subsequent gradual cooling to room temperature (Park et al. 2016). Compressive strength test is conducted as per ASTM C39/C39M-21 (2021) on cylindrical AAC specimens of 150 mm diameter and height 300 mm. Capacity of the compression test setup is 2000 kN.

The mineralogical, chemical, and microstructural analyses are conducted on powdered or hardened paste specimens to avoid interference due to the presence of aggregates in mortar and concrete. X-ray diffraction (XRD) analyses are performed using a RIGAKU Ultima X-ray Diffractometer. Rate of scanning equal to 1°/min with intervals of 0.02° is adopted over a 2θ range of 5–90°. The CuK_α X-rays are operated at 40 kV and 30 mA. Scanning electron microscopy (SEM) using an FEI Apreo setup is used for microstructural analyses. Oven-dried samples are sputtered with a gold–palladium layer of 10 nm thickness using LEICA EM ACE200 setup before the SEM–EDS analyses.

For the bond strength determination, the pull-out tests on different AAC mixes are conducted according to the IS 2770-1 (1967). The test specimens have size 100 mm × 100 mm × 100 mm containing an embedded centrally placed rebar. The test setup is shown in Fig. 9.1. The specimens are exerted to displacements of 0.01 mm/sec till they reach failure. The average bond stress (MPa), σ_{bond}, corresponding to the specimen failure load (N), P, is determined using the following equation:

$$\sigma_{bond} = \frac{P}{\pi d L},$$

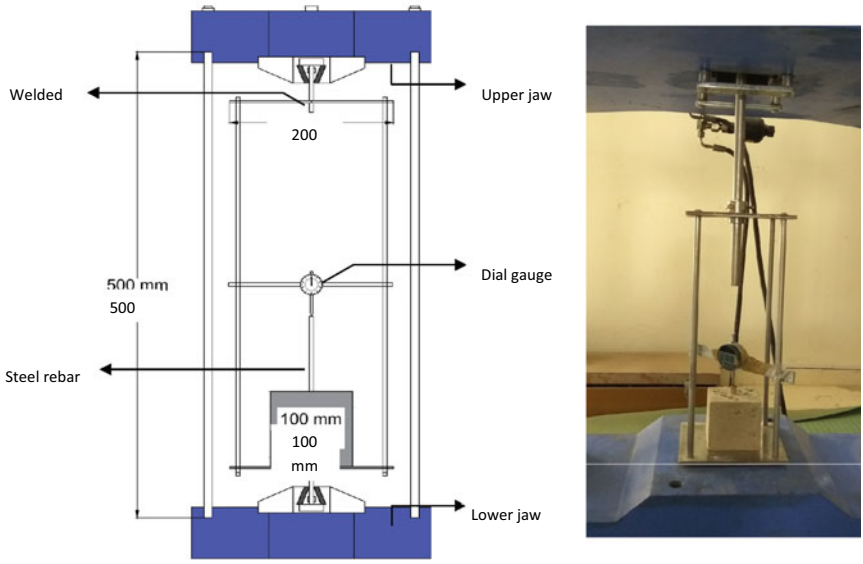

Fig. 9.1 Pull-out test apparatus

where d and L represent rebar diameter and the length of specimen parallel to the rebar, respectively (mm)In this case, $L = 100$ mm that is equal to the length of the rebar embedded inside the concrete specimen. For this study, the rebar diameter used is 12 mm. The detailed compressive and bond strength values are presented in Figs. 9.2 and 9.6, respectively.

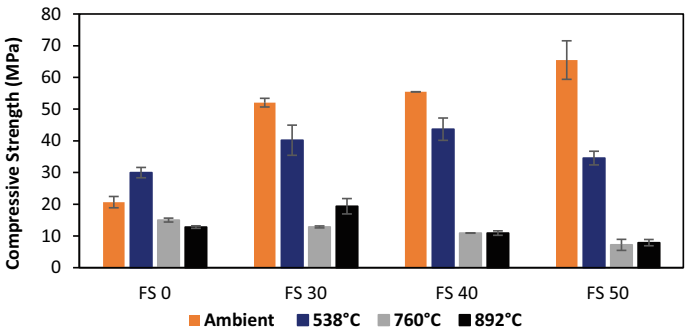

Fig. 9.2 Compressive strength results

9.3 Results

The compressive strength values for the selected exposure conditions 28 days are presented in Fig. 9.2. Ambient-cured AAC exhibits enhanced strength with increasing slag content, as reported in the literature (Kar et al. 2013). The development of C-A-S-H yields a denser microstructure and consequent pore refinement. At elevated temperatures, C-A-S-H disintegrates, leading to increased strength loss as slag proportion increases. Ambient curing results in lower fly ash reaction rates and hence lower strength development in FS 0 mixes. However, additional geopolymerization at temperatures of about 500 °C leads to an increase in strength.

Up to a temperature of 760 °C, FS 30 and FS 40 show a reduction in strength, but exposure to 892 °C leads to a relative increase in the strength. The formation of akermanite ($2CaO{\cdot}MgO{\cdot}2SiO_2$) and gehlenite ($2CaO{\cdot}Al_2O_3{\cdot}SiO_2$) crystalline phases at this elevated temperature causes an increase in the compactness of the microstructure, leading to this observation. These findings are corroborated by the diffractograms presented in Fig. 9.3 (1: analcime, 2: akermanite, 3: albite, 4: anorthite, 5: Calcite, 6: C-A-S-H, 7: gehlenite, 8: Hydrotalcite, 9: mullite, 10: nepheline, 11: quartz). FS 0 mixes exhibit a relatively lesser loss in strength at elevated temperatures due to greater pore content in the microstructure. Fly ash comprises hollow spherical particles at the molecular level, known as cenospheres. The sintering of these cenospheres at elevated temperatures leads to the formation of a highly dispersed network of pores, as observed from the micrographs presented in Fig. 9.4. This network serves as an escape route for high pore pressure, which can otherwise cause failure due to spalling or disintegration of the concrete.

Visual inspection of the specimens exposed to elevated temperatures reveals changes in superficial color due to dehydration and corresponding bond failure modes (Fig. 9.5).

Figure 9.6 presents the bond strength test values for the different AAC mixes, with the trends resembling those of compressive strength values.

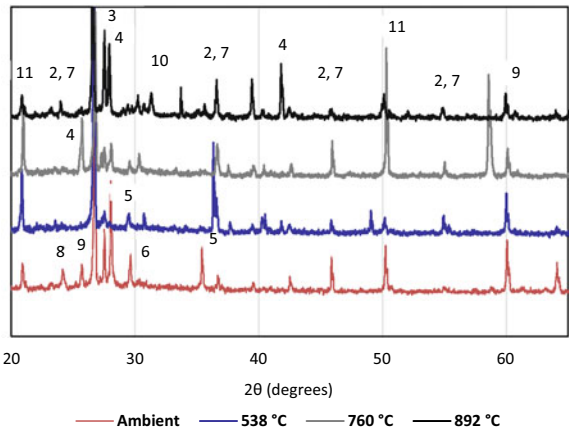

Fig. 9.3 Diffractogram for FS 50

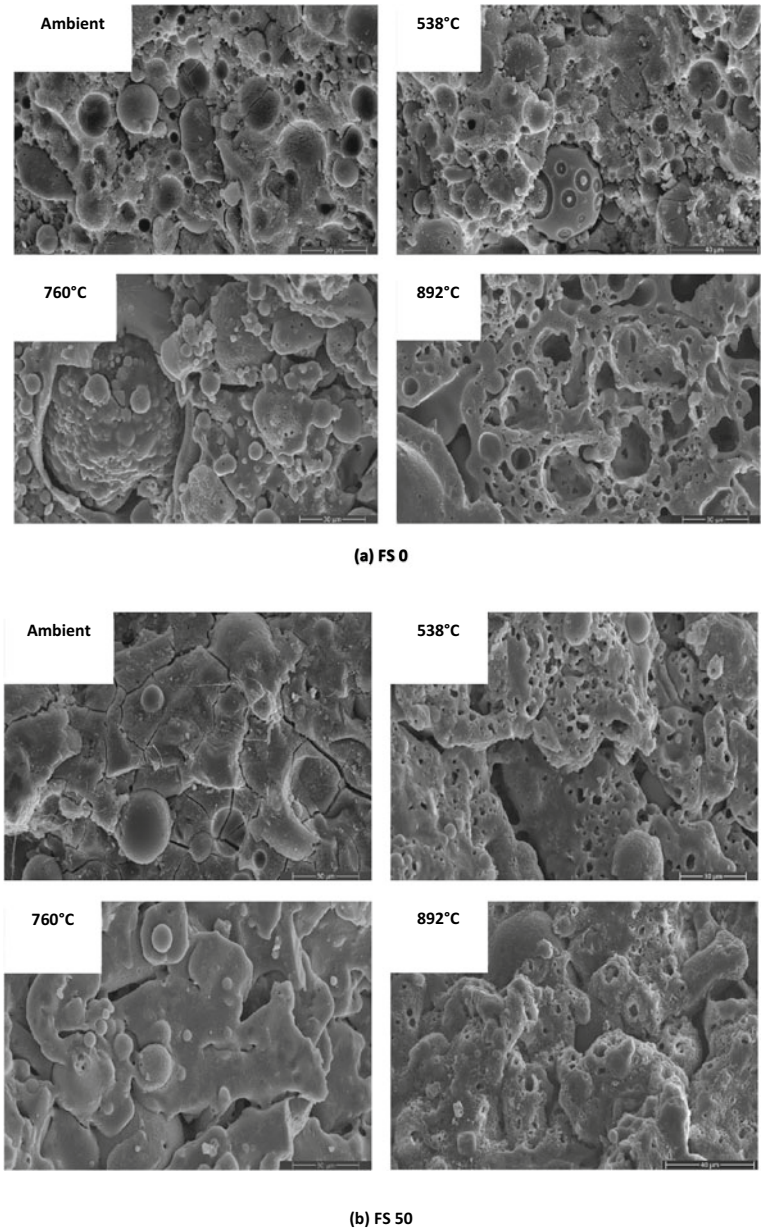

Fig. 9.4 Micrographs captured at 2500x magnification

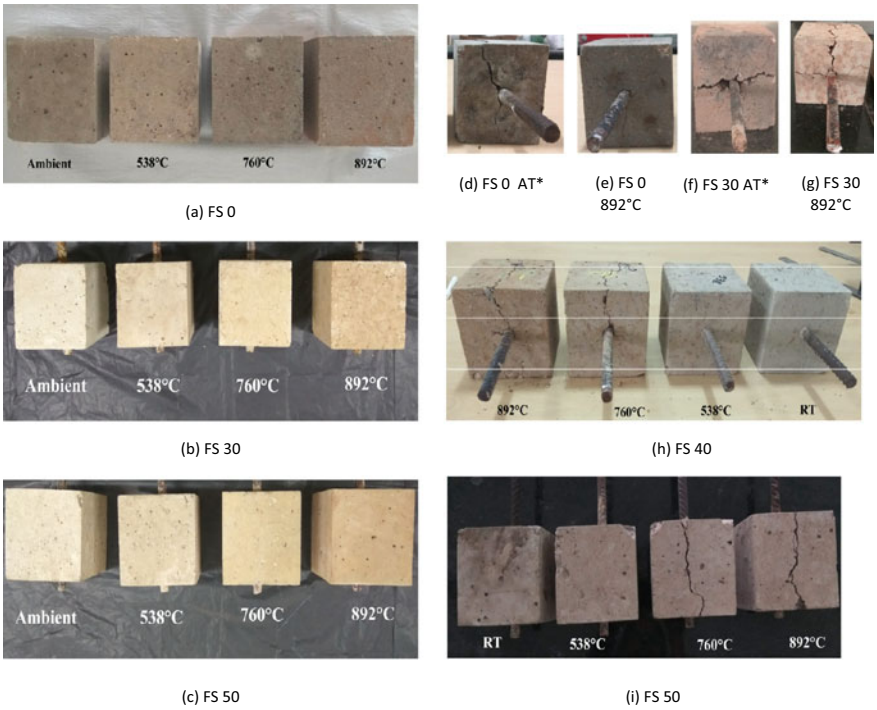

Fig. 9.5 Surface deterioration and bond failure patterns due to elevated temperature. *AT = ambient temperature

Fig. 9.6 Bond strength variation with exposure temperature

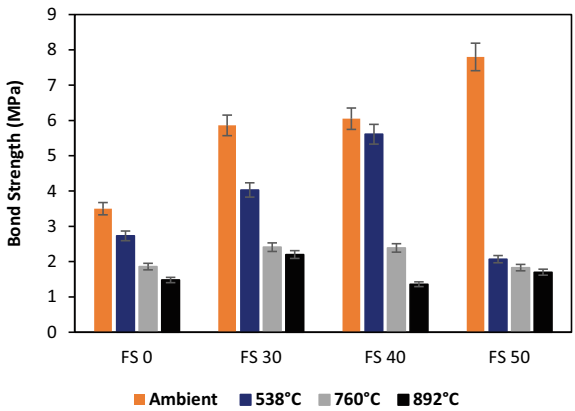

Table 9.3 Summary of observed bond failure

Mix ID	Exposure temperature	
	Ambient	Elevated
FS 0	Concrete splitting	Splitting + slippage
FS 30	Rebar slippage	Splitting + slippage
FS 40	Rebar slippage	Splitting + slippage
FS 50	Rebar slippage	Splitting + slippage

FS 50 exhibits a 122.8% higher bond strength value than FS 0, owing to a refined pore structure and a relatively homogenous and compact AAC-rebar ITZ. With increasing slag content, greater volumes of reaction products are formed, as evident from the coexistence of the C-A-S-H and N-A-S-H matrices. This greater depletion in bond strength with increasing exposure temperature and slag content occurs due to the disintegration of the C-A-S-H matrix. In FS 0, the bond between the rebars and the polymeric N-A-S-H is less prone to decomposition at higher temperatures, resulting in higher residual bond strength after exposure. The pull-out test observations presented in Fig. 9.6 are summarized in Table 9.3.

Bond failure mode depends on the splitting tensile strength of AAC, concrete cover, geometry of the rebar, and the friction between steel and concrete. Under ambient conditions, splitting and slippage are observed. At elevated temperatures, differential expansions in AAC and steel reduce inherent friction while disintegrating the concrete microstructure, leading to combined rebar slippage and concrete splitting failure.

9.4 Conclusion

1. Slag addition leads to C-A-S-H formation in addition to N-A-S-H. The formations of akermanite and gehlenite impart additional residual strength at 892 °C.
2. Slag content proportionally enhances the mechanical performance of ambient-cured AAC.
3. Rebar slippage is observed as the bond failure mode upon slag addition.
4. FS 30 is recommended as the optimal precursor combination as it shows 145% higher residual compressive strength and 29% higher bond strength than FS 50.

References

Adak D, Sarkar M, Mandal S (2017), Structural performance of nano-silica modified fly ash based geopolymer concrete. Constr Build Mater 135:430–439. https://doi.org/10.1016/j.conbuildmat.2016.12.111

ASTM C989/C989M-18a (2018) Standard specification for slag cement for use in concrete and mortars. ASTM International, West Conshohocken, PA, 2018, vol 04.02. http://www.astm.org. https://doi.org/10.1520/C0989_C0989M-18A

ASTM E119-20 (2020) Standard test methods for fire tests of building construction and materials. ASTM International, West Conshohocken, PA, 2020, vol 04.07. http://www.astm.org. https://doi.org/10.1520/E0119-20

ASTM C39/C39M-21 (2021) Standard test method for compressive strength of cylindrical concrete specimens. ASTM International, West Conshohocken, PA, 2019, vol 04.02. http://www.astm.org. https://doi.org/10.1520/C0039_C0039M-21

ASTM C618-19 (2019) Standard specification for coal fly ash and raw or calcined natural Pozzolan for use in concrete. ASTM International, West Conshohocken, PA, 2019, vol 04.02. http://www.astm.org. https://doi.org/10.1520/C0618-19

BEEINDIA (2021) Cement, bureau of energy efficiency, Government of India, Ministry of Power. https://www.beeindia.gov.in/node/166. Accessed 04 July 2021

Castel A, Foster SJ (2015) Bond strength between blended slag and Class F fly ash geopolymer concrete with steel reinforcement. Cem Concr Res 72:48–53. https://doi.org/10.1016/j.cemconres.2015.02.016

Eurostat (2020) Treatment of waste by waste category, hazardousness and waste management operations. Eurostat. http://www.data.europa.eu/data/datasets/8bxb7vunmkpy3c2mnoelw?locale=en. Accessed 04 July 2021

Fernandez-Jimenez AM, Palomo A, Lopez- C (2006) Engineering properties of alkali-activated fly ash concrete. ACI Mater J 103(2):106–112

Garside M (2021) Major countries in worldwide cement production 2010–2020. Statista. https://www.statista.com/statistics/267364/world-cement-production-by-country/. Accessed 04 July 2021

Ghazali N, Muthusamy K, Wan Ahmad S (2019) Utilization of fly ash in construction. IOP Conf Ser Mater Sci Eng 601:012023. https://doi.org/10.1088/1757-899X/601/1/012023

Glasby T, Day J, Genrich R, Aldred J (2015) EFC geopolymer concrete aircraft pavements at Brisbane West Wellcamp Airport. In: Concrete 2015 Conference, Melbourne Australia, 2015, pp 1–9. https://www.wagner.com.au/media/1512/bwwa-efc-pavements_2015.pdf. Accessed 04 July 2021

IS 2770-1 (1967) Methods of testing bond in reinforced concrete, Part 1: Pull-out test. Bureau of Indian Standards, New Delhi, India

IS 383 (2016) Coarse and fine aggregate for concrete—specification. Bureau of Indian Standards, New Delhi, India

Kar A, Halabe UB, Ray I, Unnikrishnan A (2013) Nondestructive characterizations of alkali activated fly ash and/or slag concrete. Eur Sci J 9(24):52–74. https://doi.org/10.19044/esj.2013.v9n24p%25p

Kar A (2013) Characterizations of concretes with alkali-activated binder and correlating their properties from micro-to specimen level. West Virginia University—Graduate Theses, Dissertations, and Problem Reports, 165. https://www.researchrepository.wvu.edu/etd/165

Martin A, Pastor JY, Palomo A, Jiménez AF (2015) Mechanical behaviour at high temperature of alkali activated aluminosilicates (geopolymers). Constr Build Mater 93:1188–1196. https://doi.org/10.1016/j.conbuildmat.2015.04.044

Park SM, Jang JG, Lee NK, Lee HK (2016) Physicochemical properties of binder gel in alkali-activated fly ash/slag exposed to high temperatures. Cem Concr Res 89:72–79. https://doi.org/10.1016/j.cemconres.2016.08.004

Provis JL (2018) Alkali-activated materials. Cem Concr Res 114:40–48. https://doi.org/10.1016/j.cemconres.2017.02.009

Ramagiri KK, Kar A (2019) Effect of precursor combination and elevated temperatures on the microstructure of alkali-activated binder. Indian Concr J 93(10):34–43. http://www.icjonline.com/explore_journals/2019/10

Ramagiri KK, Kar A (2020) Effect of high-temperature on the microstructure of alkali-activated binder. Mater Today Proc 28(2):1123–1129. https://doi.org/10.1016/j.matpr.2020.01.093

Ramagiri KK, Chauhan D, Gupta S, Kar A, Adak D, Mukherjee A (2021) High-temperature performance of ambient-cured alkali-activated binder concrete. Innov Infra Sol 6(2):1–11. https://doi.org/10.1007/s41062-020-00448-y

Sarker PK (2011) Bond strength of reinforcing steel embedded in fly ash-based geopolymer concrete. Mater Struct 44(5):1021–1030. https://doi.org/10.1617/s11527-010-9683-8

Yousuf A, Manzoor SO, Youssouf M, Malik ZA, Khawaja KS (2020) Fly ash: production and utilization in India-an overview. J Mater Environ Sci 11(6):911–921. http://www.jmaterenvironsci.com/Document/vol11/vol11_N6/JMES-2020-1182-Yousuf.pdf. Accessed 04 July 2021